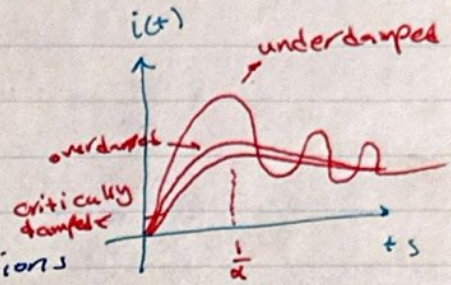



## Response in source free Series RLC

$$\begin{bmatrix} v(0) = \frac{1}{C}\int_{-\infty}^{0} i\,dt = V_0 & \text{capacitor} \\ I(0) = I_0 & \text{inductor} \end{bmatrix}$$




$$\frac{di(0)}{dt} = -\frac{1}{L}(RI_0 + V_0) \longrightarrow \text{3 types solutions}$$

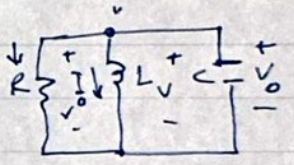
**overdamped** $i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$ $\quad \alpha > \omega_0$, $C > 4L/R^2$ $\quad s_1, s_2 = (-)$, Real

**critically damped** $i(t) = (A_2 + A_1 t)e^{-\alpha t}$ $\quad \alpha = \omega_0$, $C = \frac{4L}{R^2}$ $\quad s_1 = s_2 = -\alpha = -\frac{R}{2L}$

**underdamped** $i(t) = e^{-\alpha t}(\underbrace{B_1}_{(A_1+A_2)}\cos\omega_d t + \underbrace{B_2}_{(A_1-A_2)}\sin\omega_d t)$ $\quad \alpha < \omega_0$, $C < \frac{4L}{R^2}$ $\quad s_{1,2} = -\alpha \pm j\omega_d$

$$\alpha = \frac{R}{2L} \qquad \omega_0 = \frac{1}{\sqrt{LC}} \qquad \omega_d = \sqrt{\omega_0^2 - \alpha^2} \qquad j = \sqrt{-1}$$

## source free paralles RLC




$$i(0) = I_0 = \frac{1}{L}\int_{-\infty}^{0} v(t)\,dt \quad \text{inductor}$$

$$V(0) = V_0 \quad \text{capacitor}$$

$$\frac{d^2v}{dt^2} + \frac{1}{RC}\frac{dv}{dt} + \frac{1}{LC}v = 0$$




**overdamped** $v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$ $\quad \alpha > \omega_0$ $\quad L > 4R^2C$ $\quad s_1, s_2 = (-)$ and Real

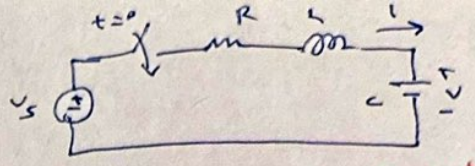
**critically damped** $v(t) = (A_1 + A_2 t)e^{-\alpha t}$ $\quad \alpha = \omega_0$ $\quad L = 4R^2C$ $\quad s_1 = s_2 =$ equal and Real

**underdamped** $v(t) = e^{-\alpha t}(A_1\cos\omega_d t + A_2\sin\omega_d t)$ $\quad \alpha < \omega_0$, $L < 4R^2C$

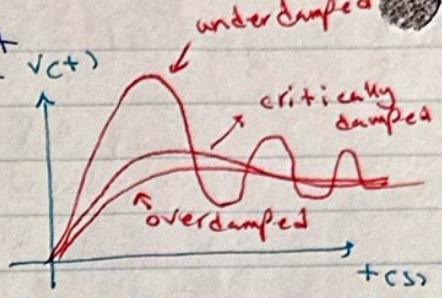
$s_{1,2} = -\alpha \pm j\omega_d$ complex

$$\frac{V_0}{R} + I_0 + C\frac{dv(0)}{dt} = 0$$

$$\frac{dv(0)}{dt} = -\frac{(V_0 + RI_0)}{RC}$$




## Step Response of a Series RLC circuit




$V_{ss} = v(\infty) = V_s$

$$L\frac{di}{dt} + Ri + v = V_s \qquad i = C\frac{dv}{dt}$$

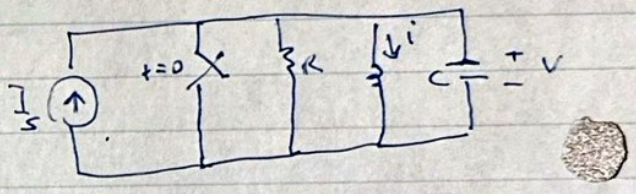
$$\frac{d^2v}{dt^2} + \frac{R}{L}\frac{dv}{dt} + \frac{v}{LC} = \frac{V_s}{LC}$$

$v(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t}$ (overdamped)

$v(t) = V_s + (A_1 + A_2 t) e^{-\alpha t}$ (critically damped)

$v(t) = V_s + [A_1 \cos \omega_d t + A_2 \sin \omega_d t)\, e^{-\alpha t}$ (underdamped)

## Step Response of a parallel RLC circuit




$$\frac{v}{R} + i + C\frac{dv}{dt} = I_s$$

$$v = L\frac{di}{dt}$$

$$\frac{d^2i}{dt^2} + \frac{1}{RC}\frac{di}{dt} + \frac{i}{LC} = \frac{I_s}{LC}$$

$i(t) = I_s + A_1 e^{s_1 t} + A_2 e^{s_2 t}$ (overdamped)

$i(t) = I_s + (A_1 + A_2 t) e^{-\alpha t}$ (critically damped)

$i(t) = I_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t)\, e^{-\alpha t}$ (underdamped)